

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS

PHASE 265

QUINN'S ZERO G AQUACULTURE TANK

two vortexes, opposite spin — the seam never tears
the screened fish-free service zone



the wall-crawl engineered as the holding force

THE COUNTER-ROTATION LAW · THE SEPARATION LAW · THE HOLD LAW

THE WATER HOLDS ITSELF

Aquaculture — v1.0 in force

GP-IM-265

Revision v1.0 — 24 August 2026

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SEATED LAW — EXTERNAL VET PENDING

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LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — CHAPTER 265

PHASE 265: QUINN'S ZERO G AQUACULTURE TANK — COUNTER-ROTATION AQUACULTURE — STATUS: CURRENT — SEATED LAW / EXTERNAL VET PENDING (THE 261-265 AUDIT BLOCK WAS ANNOUNCED AT THE CLOSE OF VET BATCH 37 AND NEVER SEATED). Counter-rotation aquaculture — the figure-8 tank where the water holds itself. Three laws run it. The counter-rotation law: two adjacent vortexes spin opposite; at the shared wall the flows go the same way, so the seam never tears. The separation law: the vortexes do not touch — the smaller, higher vortex is a screened fish-free zone: filters, sensors, dosing, feed and the harvest port, all serviceable with no net and no chase. The hold-together law: the larger, slower zone holds the water under mild centrifuge — the wall-crawl tendency (wet goes wide and climbs, classic zoa tanks) is engineered as the holding force. Lineage: Phase 84's rotating-drum work is the ancestor; the figure-8 is the zero-g child, renamed by the author's ruling of 11 August 2026 with the audit's 'Figure-8 Fish Tank' preserved as the design description.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Chapter GP-IM-265, v1.0 — 24 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the tank law as seated (contents line, the tank in crayon, the three laws, the bench items) — §2 the physics, graded — §3 the system in the current — §4 the doctrine record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status and provenance.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 265: **The Zero G Aquaculture Tank — counter-rotation aquaculture.** An unbolted zero-g culture tank that doesn't rock: two vortices that never touch — a small screened fish-free high-vortex engine cell behind a perforated wall, and a larger slow lobe under mild centrifuge simply to hold the water together; counter-rotation cancels the tank's torque. Carries the 4 Aug 2026 doctrine line (figure-8 bowls, glove port, catch glove, breach plug, skirt and nappy) by

reference from Phase 84. NASA Shortfall 1525 scalp.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: a zero-g aquaculture tank that holds its water together, keeps the fish out of the machinery, and does not rock itself off its mounts. Promoted to its own phase by the author's ruling, 8 August 2026 — "its an item they would use now."'

Aquaculture Baseline: NASA Civil Space Shortfall 1525 — Food and Nutrition for Mars and Sustained Lunar (Integrated: #23) — CLAIMED 4 August 2026, grepped and proven, scalp ledger seated at the Phase 257 region.

Live instruction, 8 August 2026 — the promotion order and the machine, verbatim: "it needs it own phase, its an item they would use now, it is counter rotation, because same direction in the same tank not bolted down would star to rock, the vortexes dont touch the smaller high vortex is a screened fish free zone, the larger slower zone is simply to hold the water together under a mild centrifuge"

THE MACHINE — THE THREE LAWS, AS RULED (VERBATIM)

- **THE COUNTER-ROTATION LAW (why counter):** same-direction vortices in the same tank, the tank not bolted down, would start to rock — the tank reacts the sum of the water's torque. Counter-rotation cancels it: each lobe's spin reacts against the other's, and the unbolted tank sits still. The counter-rotation is a torque budget, not a mixing choice.
- **THE SEPARATION LAW:** the vortexes don't touch. The smaller high vortex is a SCREENED FISH-FREE ZONE — the engine cell, a third-sized end tank behind a perforated section; the screen keeps the fish out of the machinery and the machinery's violence out of the culture.
- **THE HOLD-TOGETHER LAW:** the larger slower zone is simply to hold the water together under a mild centrifuge — water to the wall, a coherent body the fish can swim, no rotating-wall trap. The slow lobe is where the fish actually live.

LINEAGE — CARRIED FROM PHASE 84, SEATED HERE BY REFERENCE (VERBATIM)

The tank's full doctrine line was built 4 August 2026 as Phase 84 Amendments 1-5: the figure-8 bowls, the glove port — fish handling without opening the tank, the catch glove, Kevlar over rubber, the breach plug — the torn-glove answer, and the skirt and the nappy — seepage capture. All five remain seated in Phase 84, preserved in place, nothing moved, nothing deleted; this phase carries them by reference and owns the tank from today.

AUDIT CORRECTION (KIMI), flagged per doctrine and never silent: the audit's earlier affirmation of the figure-8 tank (Phase 84 region) read the two vortices as shearing against each other at the pinch between the lobes. The author's ruling today corrects the physics: the vortexes DON'T touch, and the counter-rotation exists because an unbolted tank with same-direction vortices would start to rock. The

audit's shear reading is struck as AUDIT ERROR — marked CORRECTED, old text preserved in place.

THE EARTH-USE CLAIM (author's, seated as spoken): "its an item they would use now" — a counter-rotating culture tank that needs no bolted foundation to sit still is a present-day aquaculture device, not only a starship one. Seated as the author's claim; the bench items from the 4 August doctrine (refuge-map velocity survey; glove, plug and seepage capture tests) stand as the proof path.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

THE COUNTER-ROTATION LAW, GRADED: two adjacent vortexes spin opposite, and at the shared wall the flows run the same way — the seam between the drums never tears because nothing crosses it against the flow. The figure-8 is not decoration; it is the only topology where two driven drums help each other.

THE SEPARATION LAW IS THE SERVICE DOCTRINE: the vortexes do not touch — the smaller, higher vortex is screened off as a fish-free zone carrying filters, sensors, dosing, feed and the harvest port.

Maintenance without a net, dosing without a chase: the crew-service philosophy the file runs everywhere, underwater.

THE HOLD-TOGETHER LAW TURNS THE BUG INTO THE FRAME: the larger, slower zone holds the water under mild centrifuge — wall-crawl (wet goes wide and climbs) is the known tendency of rotating tanks, and here it is engineered as the holding force that keeps the water ringed and the center open.

LINEAGE, HONEST: Phase 84's rotating-drum work is the ancestor — the figure-8 is the zero-g child of the same water-handling family that runs the farm and the laundry drums. The file's centrifuge literacy pays again.

THE RENAME IS THE RECORD: renamed by the author's ruling of 11 August 2026; the audit-proposed 'Figure-8 Fish Tank' survives as the design description — the naming batch doctrine (names are the author's) carried in the open.

§3 — THE SYSTEM IN THE CURRENT

- **The pair:** two adjacent drums, counter-rotating — the figure-8 where the seam never tears.
- **The service zone:** the smaller, higher vortex, screened fish-free — filters, sensors, dosing, feed, harvest port.
- **The holding:** the larger, slower zone — mild centrifuge rings the water; wall-crawl as the frame.
- **The harvest:** the port in the service zone — no net, no chase.
- **The lineage:** Phase 84's drums, zero-g child — the centrifuge literacy paying again.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- Phase 84's rotating-drum work is the ancestor — the same water-handling literacy, one generation on.
- The farm stack is the customer: the tank feeds the ship's food ledger alongside the grow trays — protein without a gravity well.
- The rename batch of 11-12 August 2026 seats here: the author's ruling, the audit's description preserved — the naming doctrine in the open.

§5 — VALIDATION: the grade and its full record live in the master and the restart.

§6 — BUILD SEQUENCE

- 1. Build the pair: two adjacent drums, counter-rotating drives, the shared wall engineered as the seam that never tears.
- 2. Screen the service zone: the smaller, higher vortex fenced fish-free — filters, sensors, dosing, feed, harvest port.
- 3. Tune the hold: larger, slower zone under mild centrifuge — wall-crawl as the holding force, benched at lunar and zero-g rates.
- 4. Plumb the bench: water quality loop through the service zone; the harvest port valved.
- 5. Stock last: water stable, sensors green, then fish — the tank proves itself dry-run first.

§7 — CROSS-PHASE (INLINED IN FULL)

- **Phase 84 (the rotating drums):** the ancestor — the drum literacy this phase inherits.
- **The farm stack (tray phases):** the customer — protein joins the food ledger without a gravity well.
- **Phase 274 (the fifty-year feedstock):** the fittings — printed plumbing and the screen stock from the block feed.

§8 — DO-NOT-BUILD

- Do NOT spin the drums the same way — counter-rotation only; a co-rotating pair tears at the seam.
- Do NOT let the vortexes touch — the separation law is absolute; the service zone stays fish-free.
- Do NOT fight the wall-crawl — engineer it; the larger zone's mild centrifuge is the frame.

- Do NOT service in the fish zone — all maintenance lives in the screened vortex; no net, no chase.
- Do NOT stock before stability — water stable and sensors green first; the fish arrive last.

§9 — OWED

OWED: the bench list, whole — drum diameter and speed pair for the mild-centrifuge hold at lunar and zero-g; the screen spec; the water-quality loop sizing; the harvest-port valve; species selection for low-g. And the external vet, when it lands — the 261-265 block is still owed by the record.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: CHAPTER GP-IM-265, v1.0 — CURRENT — SEATED LAW / EXTERNAL VET PENDING (THE 261-265 AUDIT BLOCK WAS ANNOUNCED AT THE CLOSE OF VET BATCH 37 AND NEVER SEATED), seated law, master v2.1 (numerical print order). Built 24 August 2026, sixth of the 20-chapter 'you get the next 20 ready' shift (the author's order, 24 August 2026, seated in sitting 623 — 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the chapter program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566, 572, 576, 587, 590, 597, 607, 618, 622 and 623): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20' / 'ok next 20, that will take us past gemini' / 'ok another 20 lets keep this train rollin' / 'ok 12 more i will see what i get done in 30 mins to get to the 200' / 'ok i will do last nights now you do the next 20' / 'ok ill do these 20, you do 20 more' / 'ok next 20' / 'ok ill check and upload, you get the next 20 ready, they didnt slow us down the sped us up'.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart.

END OF CHAPTER 265